

TECHNICAL ABSTRACT: MULTI-SOURCE CANDIDATE DATA TRANSFORMER

System Architecture & Design Review | Candidate: Engineering Intern | Target: Eightfold AI Senior Reviewers

1. System Architecture & Pipeline

The system implements a decoupled, linear data ingestion pipeline separating parsing, mapping, normalization, conflict resolution, score evaluation, and custom projection layer. Underpinned by clean design architecture, modules strictly fulfill the **Single Responsibility Principle (SRP)** to ensure scalability on production workloads.

2. Heterogeneous Parsers

- **CSV/JSON:** Read structured headers mapping variations (e.g. designation -> headline) into canonical keys.
- **PDF/TXT:** Utilizes *pdfplumber* for layout-preserved extraction (falling back to *pypdf* if corrupt) and runs optimized regex filters for structured fields.

3. Value Normalization Engine

- **Phones:** Cleans digits and standardizes to E.164 (e.g. +919876543210).
- **Dates:** Parses multiple formats (Jan 2024, 01/24, 2024) to ISO YYYY-MM.
- **Skills:** Maps lowercase tech aliases (ReactJS, React.js) to canonical labels (React).
- **Countries:** Standardizes names and codes to ISO-3166-1 alpha-2.

4. Confidence Scoring Model

- We introduce a multi-factor confidence model. Base confidences depend on ingestion reliability (Structured = 0.85; Unstructured = 0.65).
- **Consensus Boost:** Field confidence gets +0.10 for each matching value detected across sources.
 - **Normalization Penalty:** Formats failing validation (e.g. malformed phone) are penalized by -0.15.
 - **Overall Confidence:** Weighted average of all resolved fields.

5. Merge & Conflict Resolution Policies

- Data merging is resolved deterministically by candidate keys (e.g. normalized email/phone). Conflict resolution rules ensure integrity:
- **Contact Details:** *CSV* > *Resume*. Structured database exports are prioritized.
 - **Work Experience:** *Resume* > *CSV*. Text summaries from resume offer superior descriptive depth.
 - **Experience Deduplication:** Jobs at the same company and title are aligned, merging start/end dates and selecting the longest string for summaries.

6. Configurable Projection Layer

- Provides runtime customization without changing Python code. Supports:
- Field Renaming & Subsetting (using dot and array-index notation like *emails[0]*).
 - Dynamic toggles for metadata (confidence scores, provenance histories).
 - Missing value policies: *null*, *omit* (removes key), or *error* (raises validation exception).

7. Edge Case Management & Safety

- **Corrupt PDF:** Recovers by falling back to *pypdf*; errors are caught and logged, returning empty files gracefully rather than crashing.
- **Empty Fields:** Missing attributes map to null values; never 'invent' data.
- **JSON Schema:** Built-in validation blocks malformed canonical or projection configs.

8. Key Architectural Trade-offs

- **Regex vs. LLM:** Text parsing uses pre-compiled regex and heuristics. While LLMs offer higher accuracy for edge layouts, regex parsers run in milliseconds, require no network calls, and guarantee 100% deterministic results, critical for bulk indexing.